Natural Language Processing

Session 03 – Linguistic Annotations & Pattern Matching

Prof. Dr. Richard Sieg
SoSe 2026

Schedule

Date	Weekday	CW	Room	Topic
16.04.	Thursday	16	149	Introduction: The World of NLP
23.04.	Thursday	17	149	Text Processing Fundamentals
30.04.	Thursday	18	149	Linguistic Annotations & Pattern Matching
05.05.	Tuesday	19	154	Text Vectorization: From Text to Numbers ⚠️ Substitute for 14.05.
07.05.	Thursday	19	149	Text Classification
28.05.	Thursday	22	149	Language Models, Word Vectors + 🎤 Guest Lecture Ines Montani
01.06.	Monday	23	390	📝 **Exams — Master DS Students**
02.06.	Tuesday	23	154	BERT: Pre-Training & Fine-Tuning ⚠️ Substitute for 04.06.
11.06.	Thursday	24	149	The BERT Ecosystem
18.06.	Thursday	25	149	Introduction to LLMs
25.06.	Thursday	26	149	Working with LLMs
02.07.	Thursday	27	149	Ethics, Limitations & Exam Preparation
09.07.	Thursday	28	149	📝 Exams (afternoon) — Bachelor DIS Students
10.07.	Friday	28	149	📝 Exams (afternoon) — Bachelor DIS Students

Recall: The typical NLP pipeline

Data Collection	Pre-processing	Feature Extraction	Modelling & Prediction	Evaluation
Raw text, corpora, web scraping, APIs	Cleaning, stemming, tokenization, stop words	TF-IDF, BoW, word/sentence embeddings	ML model neural network, training & inference	Compare vs gold labels, precision, recall, F1

Agenda

01. Part-of-Speech Tagging

02. Dependency Parsing

03. Named Entity Recognition

04. spaCy Pattern Matching (live notebook)

01.

Part-of-Speech Tagging

What's in a spaCy token?

```
doc = nlp("She sells seashells")
print(f"{'Text':<10} {'Lemma':<10} {'POS':<10} {'Tag':<10} {'Dep':<10}")
for token in doc:
    print(f"{token.text:<10} {token.lemma_:<10} {token.pos_:<10} {token.tag_:<10} {token.dep_:<10}")

Text       Lemma      POS        Tag        Dep
She        she        PRON       PRP        nsubj
sells      sell       VERB       VBZ        ROOT
seashells  seashell   NOUN       NNS        dobj
```

What do these mean?

What Are Parts of Speech?

- Dionysius Thrax of Alexandria (c. 100 B.C.) described eight parts of speech:
  - noun, verb, pronoun, preposition, adverb, conjunction, participle, article
- Still the basis of modern NLP tagsets

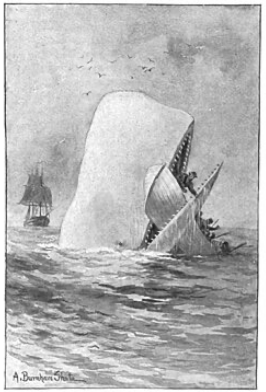
Every word in a sentence plays a syntactic role.
POS tags capture that role.

She	can	not	sell	the	two	old	houses	in	Cologne	.
PRON	AUX	PART	VERB	DET	NUM	ADJ	NOUN	ADP	PROPN	PUNCT

Why POS Tags Are Morphosyntactic, Not Semantic

You might think

- Nouns = things, Verbs = actions, Adjectives = properties
- But POS tags are about grammatical behavior, not meaning:
  - *"...the howling of the shrieking storm"*
  - howling = action → but grammatically a NOUN
  - shrieking = action → but grammatically an ADJ
- *"Police Help Dog Bite Victim"*
  - Reading 1: help = VERB, bite = NOUN
  - Reading 2: help = VERB, bite = VERB



"Both jaws, like enormous shears, bit the craft completely in twain."

—Page 510.

POS disambiguates these readings.

Word Classes

Open Classes

New words are added regularly.
Content words with lexical meaning.

Nouns (NOUN)

Describe entities and concepts

cat, algorithm, beauty

Adjectives (ADJ)

Describe properties

red, young, awesome

Verbs (VERB)

Signal events and actions

draw, provide, go

Adverbs (ADV)

Modify verbs, adjectives, or other adverbs

very, slowly, yesterday

Closed Classes

Fixed membership. Function words with grammatical purpose.

Adpositions (ADP)

Relate nouns to other parts of the sentence

in, on, by

Auxiliaries (AUX)

Add tense, aspect, mood, modality

can, should, are

Determiners (DET)

Articles, demonstratives, quantifiers

a, the, this

Pronouns (PRON)

Substitute for noun phrases

she, who, I

Particles (PART)

Infinitival to, possessive 's, negation not

's, not, (infinitive) to

The Universal Dependencies Tagset (UPOS)

	Tag	Description	Example
Open	ADJ	Adjective	red, young
	ADV	Adverb	very, slowly
	NOUN	Noun	algorithm, cat
	VERB	Verb	draw, go
	PROPN	Proper noun	IBM, Colorado
	INTJ	Interjection	oh, hello
Closed	ADP	Adposition	in, on, by
	AUX	Auxiliary	can, should
	CCONJ	Coord. conj.	and, or
	DET	Determiner	a, the, this
	NUM	Numeral	one, 2026
	PART	Particle	's, not, to
	PRON	Pronoun	she, who, I
	SCONJ	Subord. conj.	whether, because
Other	PUNCT	Punctuation	; , ()
	SYM	Symbols	$, %
	X	Other	asdf, qwfg

`token.pos_` in spaCy

The Penn Treebank Tagset (PTB)

PTB distinguishes…	Example tags	UPOS equivalent	Example
Noun singular vs. plural	`NN vs. NNS`	both NOUN	*cats/NN vs. cat/NNS*
Proper noun sing. vs. plural	`NNP vs. NNPS`	both PROPN	*London/NNP vs. Romans/NNPS*
Verb tenses & forms	`VB, VBD, VBG, VBN, VBP, VBZ`	all VERB	*run/VB, ran/VBD, running/VBG*
Adjective degree	`JJ, JJR, JJS`	all ADJ	*big/JJ, bigger/JJR, biggest/JJS*
Adverb degree	`RB, RBR, RBS`	all ADV	*fast/RB, faster/RBR, fastest/RBS*
Existential "there"	`EX`	PRON in	*There/EX is a problem*
Preposition + subord. conj.	`IN (merged)`	ADP or SCONJ	*in/IN the box; if/IN you go*

45 tags, ~3× more granular than UPOS.

The PTB tagset was designed for **English only**. It captures morphological detail that UPOS *deliberately* leaves out (UD captures those via separate attribute annotations).

`token.tag_` in spaCy

How Ambiguous is POS Tagging?

- ~85% of word types are unambiguous (always the same POS):
  - Janet → always PROPN; hesitantly → always ADV
- But the 15% ambiguous types include very common words
  - 55 to 67% of running tokens are ambiguous.

The word "back" has 5/6 possible tags:

It depends on the context.

She sat in the back seat.	ADJ
A small building in the back.	NOUN
Senators back the bill.	VERB
He backed out.	VERB
They buy back debt.	PART
I was twenty-one back then.	ADV

Training POS Taggers – Sequence Labeling

Step 1: Annotated Data (Treebanks)

- Humans manually label each word in a corpus with POS tags.
  - Penn Treebank (1993): 1M words of Wall Street Journal text, 45 tags
  - Universal Dependencies: 200+ treebanks in 100+ languages, 17 tags

Step 2: Machine Learning

- The tagger learns patterns from labelled data using these features:
  - The word itself → "the" is almost always DET
  - Surrounding words → word before "the" is usually not a verb
  - Previous POS tags → DET is usually followed by NOUN or ADJ
  - Word shape → starts with capital? contains digit?
  - Prefixes/suffixes → "-ly" → ADV, "-tion" → NOUN

Evolution of POS Taggers

Simplest possible tagger:

1. For each word, look up the most common tag in the training data.
2. If the word is unknown, guess NOUN.

This already achieves around 90% accuracy.

Model Type	Typical Accuracy (Penn Treebank)	Main Strength
Baseline (Most Frequent Tag)	**~90%**	Extremely fast; no context.
HMM (Trigram / TnT Tagger)	**94% – 95%**	Fast; uses local context (2–3 words).
CRF (Conditional Random Fields)	**96% – 97%**	Can use many "hand-crafted" features.
Bi-LSTM / Transformers	**97.5% +**	Captures long-range dependencies.

Why POS Tagging Matters

Task	How POS helps
Parsing	POS constrains possible dependency relations
Lemmatization	"better" → NOUN: "better", ADJ: "good"
NER	Proper nouns (PROPN) are candidate named entities
Text-to-speech	"lead" (NOUN: /lɛd/ vs. VERB: /liːd/)
Sentiment analysis	Adjectives carry sentiment; nouns carry topics
Machine translation	Adjective-noun reordering across languages
Information extraction	Verbs identify actions; nouns identify participants

02.

Dependency Parsing

From POS Tags to Structure

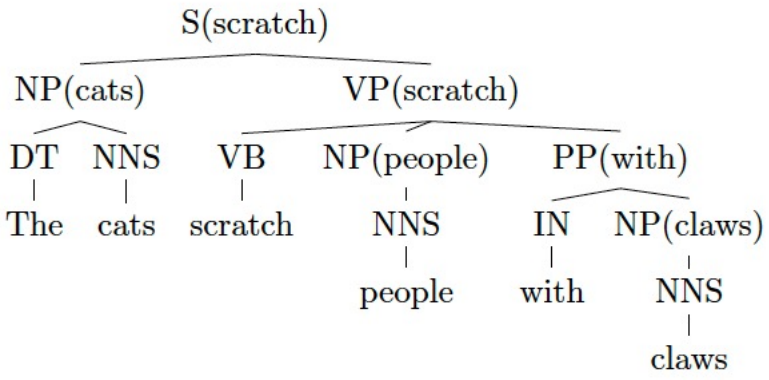
- POS tags tell us what each word is: noun, verb, adjective...
- But they don't tell us how words relate to each other.

I	saw	the	man	with	the	telescope
PRON	VERB	DET	NOUN	ADP	DET	NOUN

- Who has the telescope? The man? Or me?
- POS tags can't answer this. ***We need structure.***

What Is Dependency Parsing?

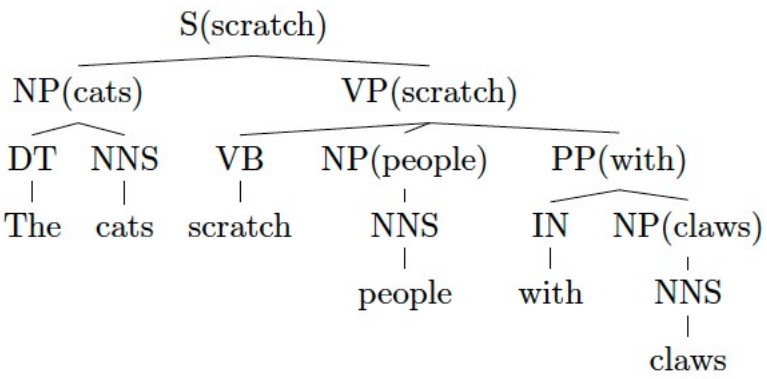
- A dependency parse describes the syntactic structure of a sentence as a set of directed binary relations between words.



- Every relation connects a head (the central word) to a dependent (the modifier).
  - *"the cats"*: *cats* is the head, *the* depends on it
  - *"cats scratch"*: *scratch* is the head, *cats* depends on it

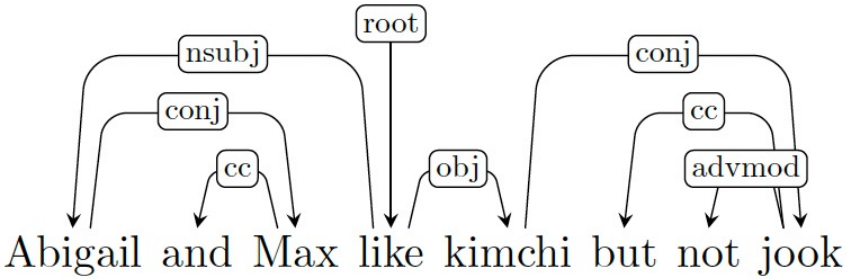
Which word is the head?

1. *The head sets the syntactic category:* nouns head noun phrases, verbs head verb phrases
2. *The head is mandatory, the dependent is optional:* "cats scratch" is a sentence, "with claws" alone is not
3. *The head determines the form of the dependent:* in German, the noun's gender determines the adjective's ending
4. *Content words first:* connect nouns, verbs, adjectives before connecting function words like determiners



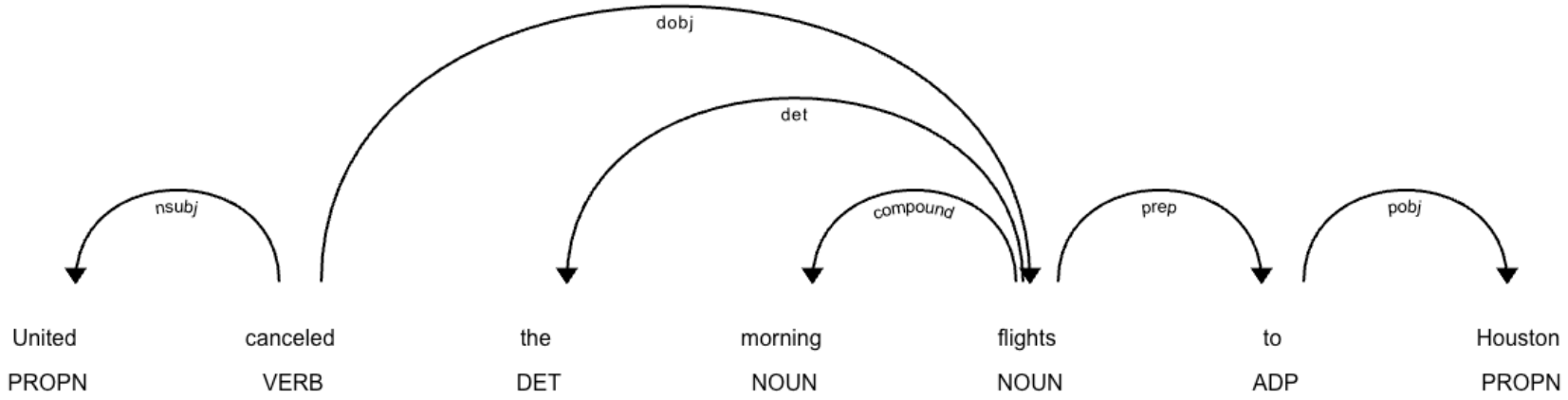
Labeled Dependencies

Arcs in the tree carry labels that describe the type of syntactic relation



Label	Meaning / Role	Example
CLAUSAL ARGUMENTS		
`nsubj`	Subject	***United*** *canceled the flight*
`obj`	Direct object	*United canceled* ***the flight***
`iobj`	Indirect object	*We booked* ***her*** *the flight*
`ccomp`	Clausal complement	*He said* ***he was tired***
MODIFIER RELATIONS		
`amod`	Adjectival modifier	*the* ***cheapest*** *flight*
`nmod`	Nominal modifier	*flight* ***to Houston***
`advmod`	Adverbial modifier	***not*** *jook / runs* ***quickly***
`det`	Determiner	***the*** *flight*
`case`	Case marker / prep.	*flight* ***to*** *Houston*
`compound`	Compound noun	***morning*** *flight*
OTHER		
`conj`	Conjunct	*Abigail and* ***Max***
`cc`	Coordinating conj.	*Abigail* ***and*** *Max*
`ROOT`	Root of sentence	(main verb)

Labeled Dependencies



Example: "Grammarly for Instructions"

For my work at how.fm, I built a system to give Grammarly like feedback about instruction texts following some rules using POS and dependency parsing e.g.

- One action should have only one target
- Only one action per instruction step
- Actions must be in second person and present tense
- No passive verbs (Germans love these!)
- No nominalization (“das Heben” - Germans love these even more!)
- Preconditions must be named first (“When the light turns green, push…”)

DIE BESTÄTIGUNG
ERFOLGT DURCH DIE TASTE
imgflip.com

03.

Named Entity Recognition

From Structure to (some degree of) Meaning

- POS tags tell us what each word is (noun, verb, adjective...).
- Dependency parsing tells us how words relate (subject, object, modifier...).
- But neither tells us what things are being talked about:

"Marie Curie moved to Paris in 1891 to study at the University of Paris."

- Who is this about? Where? When? What institution?
- ***Named Entity Recognition (NER)*** finds and classifies these references in text.

What Are Named Entities?

- Anything that can be referred to with a proper name
- Often multi-word phrases *"New York City"*, *"Werder Bremen"*

Type	Tag	Examples
Person	**PER**	Marie Curie, Steve Jobs
Organization	**ORG**	United Airlines, TH Köln
Location	**LOC**	the Rhine, Mount Everest
Geo-Political Entity	**GPE**	Germany, the EU

- The term is extended beyond proper names to include

Type	Tag	Examples
Date	**DATE**	Friday, May 14 1864
Time	**TIME**	3pm, midnight
Money	**MONEY**	$6, €1.2 million
Percent	**PERCENT**	28%, three quarters

spaCy Entity Types

TYPE	DESCRIPTION
PERSON	People, including fictional.
NORP	Nationalities or religious or political groups.
FAC	Buildings, airports, highways, bridges, etc.
ORG	Companies, agencies, institutions, etc.
GPE	Countries, cities, states.
LOC	Non-GPE locations, mountain ranges, bodies of water.
PRODUCT	Objects, vehicles, foods, etc. (Not services.)
EVENT	Named hurricanes, battles, wars, sports events, etc.
WORK_OF_ART	Titles of books, songs, etc.
LAW	Named documents made into laws.
LANGUAGE	Any named language.
DATE	Absolute or relative dates or periods.
TIME	Times smaller than a day.
PERCENT	Percentage, including "%".
MONEY	Monetary values, including unit.
QUANTITY	Measurements, as of weight or distance.
ORDINAL	"first", "second", etc.
CARDINAL	Numerals that do not fall under another type.

NER is difficult

Problem 1: Segmentation - In NER, we must find span boundaries:

- Where does the entity start?
- Where does it end?
- Is "New" part of "New York City" or just an adjective?

Problem 2: Type ambiguity - The same name can refer to different entity types:

`[PER Washington] was born on Feb. 22nd, 1732.`

`[ORG Washington] went up 2 games to 1 in the four-game series.`

`Starmer arrived in [LOC Washington] for his last state visit.`

`In June, [GPE Washington] passed a primary seatbelt law.`

BIO Tagging: Spans as Sequence Labels

The trick: convert the span-finding problem into a per-token labeling problem (just like POS tagging).

B = Begins a span, **I** = Inside a span, **O** = Outside any span

Marie	Curie	entered	the	University	of	Paris	in	1891
B-PER	I-PER	O	O	B-ORG	I-ORG	I-ORG	O	B-DATE

Sometimes only **I** and **O** and sometimes with additional **E**nd and **S**ingle.

Why NER Matters

Application	How NER helps
Sentiment analysis	What entity is the sentiment about? "I love Apple but hate Google"
Question answering	Answer "Where was Marie Curie born?" by finding PERSON + GPE entities
Information extraction	Extract facts: [ORG United Airlines] increased fares by [MONEY $6]
Search & recommendation	Link mentions of [GPE Cologne] to a knowledge base entry
News analysis	Track which [PER politicians] and [ORG companies] appear together
Biomedical NLP	Find [PROTEIN glucocorticoid receptors] in research abstracts

NER is often the **first step** in turning unstructured text into structured data.

When spaCy NER Gets It Wrong

False Negatives

Domain-specific terms the model never saw: "Treblinka", "CRISPR-Cas9"

False Positives

Common words misidentified: "Will" tagged as PERSON

Type Confusion

"Washington" → PER or GPE or ORG?
"Apple" → ORG or fruit?

What can you do?

1. Use a larger spaCy model (`en_core_web_lg` or `en_core_web_trf`)
2. Add rule-based patterns to cover known entities → EntityRuler (coming up next!)
3. Fine-tune the NER model on your domain data

Pattern Matching in marimo notebook